

PDE Project Report:
Indoor Wireless Link Design with Helmholtz PDEs

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Author Contributions

Table 1: Summary of contributions by each group member.

Name	Contribution
Zyad Bougerfaoui	Project search and proposal idea, literature review, Helmholtz PDE research and derivation with assumptions, MATLAB simulations for signal propagation, parametric analysis and visual proof, contributions to FEM mesh analysis, engineering design of the TX/RX approach, and limitations.
Abdallah Taha	Literature review, FEM setup conditions and mesh analysis, numerical solution, analytical solution, engineering design of the TX/RX approach, and limitations.
Muhammad Ibrahim	FEM setup conditions and mesh analysis, numerical solution for the analytical solution, MATLAB code simulating the different transmitter positions, and subsections on PDE model and geometry construction, analytical solution, reflective walls, partially absorbing walls, threshold value, and optimal location and future improvements.
Youssef Andraws	Project search and proposal idea, literature review and Helmholtz PDE research, mathematical modeling, Helmholtz derivation, initial assumptions, boundary conditions, and PDE solution, engineering design of TX/RX with limitations, and FEM analysis contributions.

1 Introduction

Indoor wireless links power WiFi, cellular, and IoT devices, but inside real rooms waves reflect from walls, furniture, and metal surfaces, creating hot spots and dead zones that simple path-loss formulas cannot predict. This project tackles the design question: where should a single transmitter be placed so that a target desk region receives reliable coverage with minimal dead spots? By reducing Maxwell's equations in air at a single tone to a 2D Helmholtz PDE on a realistic room slice, and imposing physically meaningful boundary conditions for metal objects and partially reflective walls, the model produces detailed field magnitude maps over the desk area. Comparing candidate transmitter locations by the fraction of the desk area above a chosen field threshold turns a vague placement problem into a clear, physics-based optimization problem, which enables better coverage, fewer transmissions, and a principled path towards efficient wireless design.

2 Part I: Transmitter Design

In this part, we focus on the transmitter design details. Realistically, we must consider certain aspects of the transmitter's design, including the antenna length, the path loss model for the system, and the modulation scheme through which information is transmitted and received between the transmitter and the receiver, namely the WiFi router and the desk region.

2.1 Antenna Length

We first determine the antenna length of the router or transmitter. Antenna lengths are selected based on the frequency at which the signal is transmitted, and it is a function of the wavelength:

$$\lambda = \frac{c}{f}, \quad \ell = \frac{\lambda}{4}. \quad (1)$$

Our carrier frequency is 2.4 GHz, which makes the wavelength approximately 3.1 cm, matching the antenna lengths of WiFi routers that send data at this range.

2.2 Data Modulation Scheme

When transmitting and receiving data, there are two general schemes: amplitude modulation and digital modulation. Both strategies are essential in data capture and recovery at the transmitter and receiver sides. However, since WiFi routers rely on digital modulation schemes, we focus on them. More specifically, one of the common digital modulation schemes for WiFi routers is Quadrature Amplitude Modulation (QAM).

2.2.1 Digital Modulation

There are two types of digital modulation techniques: digital baseband modulation and digital bandpass modulation. In digital modulation, analog or digital inputs are acceptable. However, if the signal is analog, then it becomes converted to a digital signal through an analog-to-digital converter. To perform analog-to-digital conversion, we sample the signal at multiples of the sample period T_s ,

and T_s is selected such that the sampling frequency is at least twice the signal's bandwidth, which is the Nyquist criterion. Once sampling is complete, we quantize the signal amplitude, dividing the amplitude into equal groups and mapping it to a specific value. Lastly, the quantized levels are mapped to a specific stream of binary bits, which is referred to as encoding. Generally, in both digital modulation techniques, the modulation order M is the number of symbols that can be transmitted, which is determined by the number of bits mapped per symbol:

$$2^k = M. \quad (2)$$

2.2.2 Quadrature Amplitude Modulation (QAM)

For this project, we use a digital bandpass modulation technique used by WiFi routers called Quadrature Amplitude Modulation (QAM). QAM allows the transmission of information through two orthogonal carriers, sine and cosine, at a carrier frequency f_c . Essentially, the bit stream is mapped into complex-valued symbols, which, when decomposed, are broken into their in-phase and quadrature components:

$$s(t) = I(t) + jQ(t). \quad (3)$$

The amplitudes for I and Q are chosen from a finite set of values. The I and Q values of each symbol are mapped into a constellation diagram, which maps the symbol based on its in-phase amplitude values on the real axis and quadrature amplitude values on the imaginary axis. Each point on the constellation plane corresponds to a different possible mapping of the bits to symbols.

Once the mapping is done, we perform pulse shaping on both the I and Q components, with each pulse being periodic at a period of T_s :

$$I(t) = \sum_k I_k p(t - kT_s), \quad (4)$$

$$Q(t) = \sum_k Q_k p(t - kT_s), \quad (5)$$

$$s(t) = \sum_k a_k p(t - kT_s). \quad (6)$$

Lastly, we perform a frequency upconversion to make it a bandpass signal to transmit it, which is done by modulating the I component with a cosine and the Q component with a sine, both oscillating at the same frequency f_c :

$$s(t) = \Re\{s(t)e^{j2\pi f_c t}\}, \quad (7)$$

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t). \quad (8)$$

When passing through the channel, the signal becomes subject to noise, which distorts the signal. At the receiver, we reconstruct the I and Q components of the complex-valued signal by mixing the received signal with $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$ and then applying a low-pass filter to obtain estimates of $I(t)$ and $Q(t)$. These are estimates, not perfect recoveries, because of channel noise. Once the I and Q components are recovered, we pass them through a matched filter, whose output we sample at multiples of the symbol period with a detector to obtain noisy approximations of the original amplitudes. A decision rule then finds the received symbol with minimum Euclidean distance:

$$\hat{a}_k = \arg \min_{a_m} |y_k - a_m|. \quad (9)$$

Based on the corresponding symbol estimate, we demap the symbol to obtain the corresponding bits. Repeating this process for the received symbols gives the original stream of transmitted data.

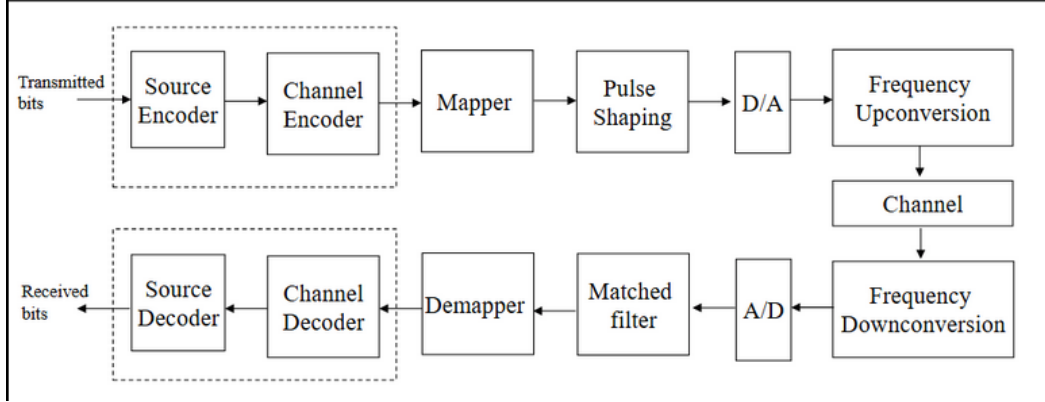


Figure 1: Schematic for Quadrature Amplitude Modulation (QAM).

3 Part II: Important Assumptions for Our System and Their Reasoning

- **No free charge in the air.** In a normal room, the air is basically neutral, meaning that there are no free clouds of charge floating in the air. Instead, any significant charge tends to sit on conducting surfaces, such as metal objects, antennas, and cables, not in the air volume. Thus, it is realistic to assume that there is no charge in the air.
- **Linear, homogeneous, and isotropic medium.** At the carrier frequency of the transmitter, which is in the GHz range, air behaves extremely close to free space. This leads to no noticeable nonlinear effects, no strong spatial variation of permittivity or permeability inside the room volume, and no preferred direction of propagation. Generally, treating air with constant scalar values ϵ and μ is standard RF practice for indoor propagation modeling.
- **Time-harmonic, single-frequency operation.** Assuming time-harmonic operation lets us replace the full time-dependent wave equation with a simpler spatial Helmholtz equation. Because the transmitted signal is modulated onto a sinusoidal carrier with frequency f_c , we can express it in complex baseband form and study the modulation in the complex plane. Most communication signals are bandlimited around f_c , so the wavenumber changes only slightly throughout the bandwidth, giving essentially the same Helmholtz problem across those frequencies. As a result, the spatial pattern is essentially the same throughout the signal's bandwidth.
- **2D horizontal slice at desk height.** Since we are finding coverage strength at a desk, we are interested in a small vertical region around desk height, not the whole room. The vertical variation is limited because most devices are placed on the desk, but the horizontal variation is much greater because we are capturing the width of the room, which leads to multipath effects due to the presence of walls. Thus, the main geometric pattern of wave propagation is more visible in the plane at desk height, allowing us to simplify the problem to 2 spatial dimensions instead of three.

- **Only the z -component of the electric field is necessary.** Since we selected a horizontal 2D slice of the room, we choose an orientation for the electromagnetic waves. There are two possible modes: transverse magnetic, where the electric field is in the z -direction and the magnetic field is in the x, y plane, and transverse electric, where the magnetic field is in the z -direction and the electric field is in the x, y plane. Since we assess coverage strength with the electric field magnitude intensity, we select the transverse magnetic mode. Because the medium is homogeneous, linear, and isotropic, the TE and TM modes decouple.
- **The transmission through the system is noiseless.** To simplify the calculations of the optimal transmitter location, we assume that the channel through which transmission is performed is noiseless, meaning that any transmitted signal is perfectly received due to a lack of channel distortion.

For the TM case, the scalar Helmholtz equation is written in terms of the z -component of the electric field:

$$\nabla^2 E_z + k^2 E_z = 0. \quad (10)$$

4 Part III: Helmholtz Equation Derivation

Let us discuss the derivation of the Helmholtz equation. We start from Maxwell's equations:

$$\nabla \cdot \mathbf{D} = \rho, \quad (11)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (12)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (13)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \quad (14)$$

To simplify the equations above, we use the assumptions imposed on the system. Initially, inside the room and away from the transmitter's antenna, there are no free charges floating in the air. Thus,

$$\rho = 0, \quad (15)$$

which implies

$$\nabla \cdot \mathbf{D} = 0. \quad (16)$$

Using linearity, homogeneity, and isotropy, we write the constitutive relation

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad (17)$$

and since ε is constant, we obtain

$$\nabla \cdot \mathbf{E} = 0. \quad (18)$$

Next, we assume there are no volume current sources in the air, so

$$\mathbf{J} = 0. \quad (19)$$

Substituting into Ampère-Maxwell's law gives

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} = \varepsilon \frac{\partial \mathbf{E}}{\partial t}. \quad (20)$$

Using $\mathbf{B} = \mu\mathbf{H}$, Faraday's law becomes

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}. \quad (21)$$

Take the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}). \quad (22)$$

Using the vector identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}, \quad (23)$$

and $\nabla \cdot \mathbf{E} = 0$, we obtain

$$-\nabla^2 \mathbf{E} = -\mu \frac{\partial}{\partial t} \left(\varepsilon \frac{\partial \mathbf{E}}{\partial t} \right). \quad (24)$$

Assuming constant ε ,

$$\nabla^2 \mathbf{E} = \mu\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (25)$$

Now we reduce this to a scalar wave equation. For the 2D horizontal slice, we assume the field has only a z -component:

$$\mathbf{E}(x, y, t) = E_z(x, y, t) \hat{\mathbf{z}}. \quad (26)$$

Thus,

$$\nabla^2 E_z = \mu\varepsilon \frac{\partial^2 E_z}{\partial t^2}. \quad (27)$$

Assuming time-harmonic fields at angular frequency $\omega = 2\pi f_c$,

$$E_z(x, y, t) = \Re\{u(x, y)e^{j\omega t}\}. \quad (28)$$

Substitute into the wave equation. The spatial operator acts only on $u(x, y)$, and the time derivatives act on the exponential:

$$\nabla^2 (u(x, y)e^{j\omega t}) = (\nabla^2 u(x, y))e^{j\omega t}, \quad (29)$$

$$\frac{\partial^2}{\partial t^2} (u(x, y)e^{j\omega t}) = -\omega^2 u(x, y)e^{j\omega t}. \quad (30)$$

The wave equation becomes

$$[\nabla^2 u(x, y) + \mu\varepsilon\omega^2 u(x, y)]e^{j\omega t} = 0. \quad (31)$$

Since $e^{j\omega t} \neq 0$, we obtain the Helmholtz equation

$$\nabla^2 u(x, y) + k^2 u(x, y) = 0, \quad (32)$$

where

$$k^2 = \mu\varepsilon\omega^2 = \left(\frac{\omega}{c}\right)^2, \quad c = \frac{1}{\sqrt{\mu\varepsilon}}. \quad (33)$$

$$\begin{aligned}
\nabla \cdot E &= \frac{\rho}{\epsilon_0} & (1) & \text{Gauss' law} \\
\nabla \cdot B &= 0 & (2) & \text{Magnetic monopoles} \\
\nabla \times E &= -\frac{\partial B}{\partial t} & (3) & \text{Faraday's law} \\
\nabla \times H &= J + \frac{\partial D}{\partial t} & (4) & \text{Ampere-Maxwell law}
\end{aligned}$$

Figure 2: Maxwell's equations.

5 Part IV: Physical Interpretation of the Helmholtz Equation

We derived the Helmholtz equation from our boundary conditions imposed on the system:

$$\nabla^2 u(x, y) + k^2 u(x, y) = 0. \quad (34)$$

We solve this equation over a 2D region of the room, which is a horizontal slice at desk height. The solution $u(x, y)$ is the z -component of the electric field at that point on the plane, and the parameter that we adjust is the wavenumber k , defined in terms of wavelength by

$$k = \frac{2\pi}{\lambda}. \quad (35)$$

Essentially, this is the time-harmonic version of the wave equation. The equation states that the electric field's z -component is constrained by its curvature, described by the Laplacian, and its oscillation, described by the term containing the wavenumber. This statement holds for waves propagating in a bounded region, since they experience continuous interference with themselves. The boundary and room conditions enforce phase shifts and reflections, while the transmitter boundary injects energy. The Helmholtz solution is the z -component of the electric field at any spatial point within the horizontal slice at steady state. In our wireless link, the bright and dead spots are determined by the magnitude of the spatial pattern of the electric field's z -component, $|u(x, y)|$, which arises through constructive and destructive interference. When we change the transmitter's position, we change the boundary condition that feeds the Helmholtz equation, so the entire wave spatial pattern in the 2D room slice changes, altering dead-zone and bright-spot locations along with the average field level at desk height. Therefore, optimizing the transmitter's location is like finding the boundary position where the Helmholtz solution has the largest coverage strength at the desk height.

6 Part V: Solution Method to the Design Problem

6.1 1. Derivation of the Helmholtz Equation from Maxwell's Equations

We have already shown how, through the conditions set on our system, we derive the Helmholtz equation from Maxwell's equations by reducing the equations to the wave equation and then converting it to a scalar version, imposing the time-harmonic condition, and deriving the wavenumber:

$$\nabla^2 u(x, y) + k^2 u(x, y) = 0. \quad (36)$$

6.2 2. Construction of the Room and its 2D Slice

We represent the space over which we are solving the Helmholtz equation as a 2D horizontal slice of the room at desk height. The outer boundary of this domain follows the floor, wall, and any major structural features that affect propagation. Within this 2D domain, we mark a subdomain representing the desk, where we evaluate the field strength to obtain the optimal transmitter location. To add more realism, we include a metal cabinet on the side of the room to create obstacles in transmission and allow for reflections and a more realistic spatial wave pattern.

6.3 3. Impose the Boundary Conditions on the System

We impose boundary conditions on the spatial wave pattern $u(x, y)$ measured along the horizontal slice. For perfectly conductive objects, such as the metal cabinet, the tangential electric field must vanish. In the scalar TM model, this is enforced by setting the field to zero at these boundaries, which gives a Dirichlet condition $u = 0$ and makes these surfaces perfect reflectors. For the room walls made from conventional building materials, we want the waves to be partially, not perfectly, reflected; to do so, we apply an impedance boundary, or Robin, condition:

$$\mathbf{n} \cdot \nabla u(x, y) + j\beta u(x, y) = 0. \quad (37)$$

Here, $\mathbf{n}$ is the outward unit normal vector and β is a parameter representing the wall's effective surface impedance, where larger β leads to more reflection.

We also need a source that drives the electric field within the room to provide the signal transmission. This is modeled as a volumetric Gaussian source:

$$f(x, y) = S_0 \exp\left(-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2}\right). \quad (38)$$

The Helmholtz equation with forcing becomes

$$\nabla^2 u(x, y) + k^2 u(x, y) = f(x, y). \quad (39)$$

This forcing term is different for each transmitter location.

6.4 4. Discretize the Domain with a Wavelength-Resolving Mesh

To solve the Helmholtz equation numerically, we discretize the 2D room slice using a finite element mesh. The mesh must be fine enough to resolve the carrier wavelength in air. A standard rule of thumb is to have on the order of 6 to 10 elements per wavelength in each spatial direction, meaning the characteristic element size should be a fraction of λ . We generate a mesh over the full room domain with this resolution, paying attention to the desk and the vicinity of the transmitter, where accurate representation of field variations is especially important. The finite element method approximates the wave spatial pattern $u(x, y)$ by values at mesh nodes and converts the continuous Helmholtz equation and boundary conditions into a large but sparse algebraic system.

6.5 5. Solve the Helmholtz Equation for Each Candidate Location

With the physics, geometry, and boundary conditions set, we solve the scalar Helmholtz equation for a given transmitter location. For a specific router position, we assign the driven boundary

condition to the small edge that represents the antenna, while keeping all other coefficients and boundaries fixed. Solving the finite element system yields a complex-valued field $u(x, y)$ defined at mesh nodes, representing the steady-state spatial distribution of the chosen field component at the carrier frequency. This process is repeated for each candidate transmitter location, resulting in a set of field maps, one per candidate.

6.6 6. Evaluate a Coverage Metric Over the Desk Region

We turn each field map into a quantitative measure of performance by defining a coverage strength metric in a region surrounding the desk. First, we choose a field magnitude threshold $|u_{th}|$ corresponding to the minimum acceptable field strength for reliable reception at the desk. For each candidate solution, we evaluate the magnitude $|u(x, y)|$ over a grid of points covering the desk area and compare it to the threshold. The coverage metric is then computed as the fraction of the desk area where $|u(x, y)| \geq |u_{th}|$. In practice, this is approximated by counting how many grid points on the desk exceed the threshold and dividing by the total number of grid points there. This single number summarizes how much of the desk is well covered for a given transmitter position.

6.7 7. Comparison of Candidate Locations and Selection of Optimal Transmitter Location

Finally, once we have found the coverage metrics across the different transmitter locations, we select the best transmitter location. For each candidate location, we compute the coverage fraction over the desk region, and the candidate with the highest coverage fraction is chosen as the optimal transmitter location. In addition to the numerical score, we visually inspect the corresponding field map to ensure there are no problematic deep nulls in practically important regions, for example where a laptop is most likely to be placed. This completes a consistent workflow: starting from Maxwell's equations, we derive a scalar Helmholtz model tailored to a 2D room slice, impose physically meaningful boundaries, solve for the field using finite elements within MATLAB's PDE environment, and evaluate a coverage-based objective to choose the transmitter placement that best serves the desk area.

6.8 8. Computational Approach and Engineering Solution

We model indoor WiFi coverage by solving the scalar Helmholtz equation for the z -directed electric field on a 2D slice of a 5 by 4 m room with a cabinet cutout, desk region, linear isotropic air, mixed absorbing and perfectly conducting boundaries, and a Gaussian source router at one of four candidate positions. Because this irregular domain, mixed boundary conditions, and localized source make an analytic solution unrealistic, we use the finite element method in MATLAB's PDE Toolbox, discretizing the domain into approximately 600,000 second-order triangular elements, solving the resulting sparse linear system for the nodal field, and interpolating to obtain smooth complex field magnitude maps showing constructive and destructive interference. We then convert these fields into an engineering metric by normalizing the magnitude, defining a 10% coverage threshold relative to the maximum, and computing for each router location the fraction of desk points above the threshold. The best performance, around 40% coverage, comes from a central placement at the coordinates (2.5 m, 2 m), while corner or wall placements suffer from cabinet shadowing and deeper nulls. A refined standing-wave analysis for the best position shows peaks

and dips spaced at roughly half a wavelength, confirming that the mesh and boundary conditions capture the correct physics and explaining why full desk coverage is unattainable with a single access point in this geometry.

7 Part VI: MATLAB Code Simulating the Different Transmitter Positions

7.1 PDE Model and Geometry Construction

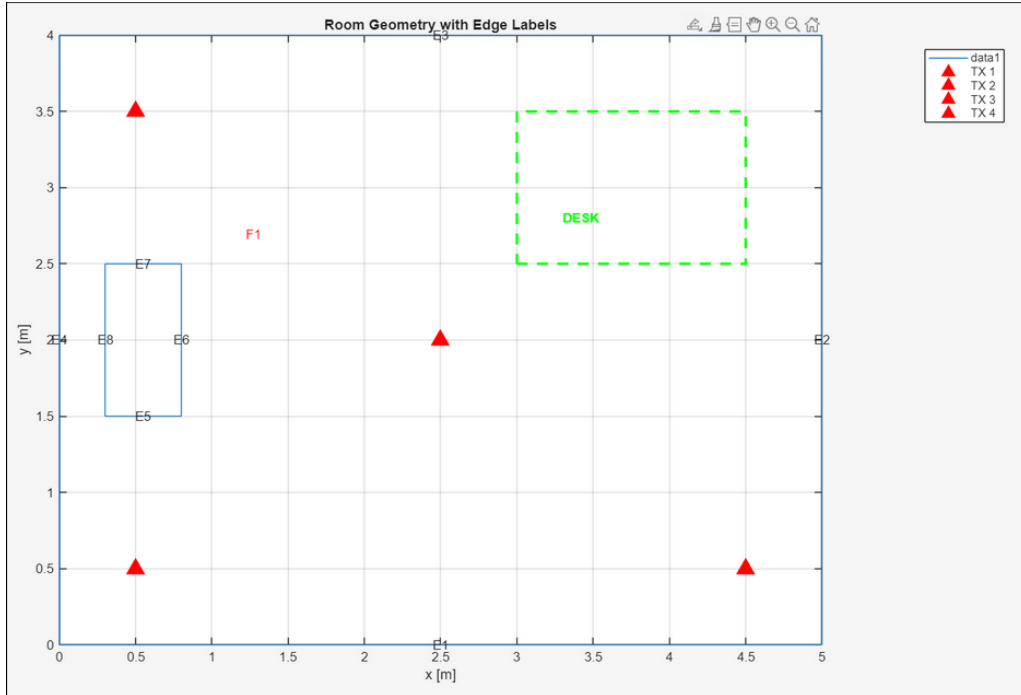


Figure 3: Domain boundaries. The 5 by 4 m rectangle represents a horizontal slice of the room at desk height, with a metal cabinet cutout at the left and a green dashed rectangle marking the desk region where signal quality is evaluated. Four red triangles mark the candidate router locations.

The 5 by 4 m rectangle represents a horizontal slice of the room at desk height, with a metal cabinet cutout at the left and a green dashed rectangle marking the desk region where signal quality is evaluated. Four red triangles mark the candidate router locations, and the outer boundary segments act as the room's walls, while the interior is the air-filled region where the scalar Helmholtz equation is solved. A perfectly conductive boundary condition is imposed on the cabinet, forcing the field to vanish and creating strong reflections, while the outer walls are partially reflective and absorbing. This geometry plot therefore encodes both the computational domain and the physical meaning of each boundary segment in the PDE model. The domain is then discretized into a triangular finite element mesh with element sizes chosen to give about fifteen elements per 12.5 cm wavelength at 2.4 GHz, allowing the oscillatory field to be resolved accurately. Quadratic triangular elements with midside nodes conform to the irregular cabinet shape, store the unknown field at each node, and enforce the PDE in a weak sense over each triangle, forming the bridge between the ideal

continuous room and the large finite linear system that is solved.

7.2 Analytical Solution

In the analytical case, the room is replaced by ideal free space: a homogeneous, unbounded medium with no reflections, absorption, or obstacles. The time-harmonic Helmholtz problem is solved with a purely outgoing radiation condition, and the room is just a viewing window. The router is placed at the center of the desk, and the electric field depends only on distance from this point, decreasing smoothly and monotonically with no standing waves or interference. The transmitter is deliberately kept at this central location because the analytical model needs a clean and symmetric field that can be written in closed form without any numerical complications. To compare this with the FEM method, this free-space field is sampled on a dense grid over the room, normalized by the room maximum, and visualized as a radially symmetric magnitude map around the router. Defining a coverage threshold in terms of normalized magnitude, about two-thirds of the desk is covered for a strict 1 percent cutoff, with the central region being strongly covered and the corners being weaker. This ideal solution acts as an upper bound for achievable coverage because it deliberately ignores reflections from walls, cabinet shadowing, or destructive interference. In contrast, the realistic FEM model shows how these mechanisms create hot spots, dead zones, and strong sensitivity to router position along with boundary conditions, while the analytical model represents the best-case pattern under perfectly benign propagation.

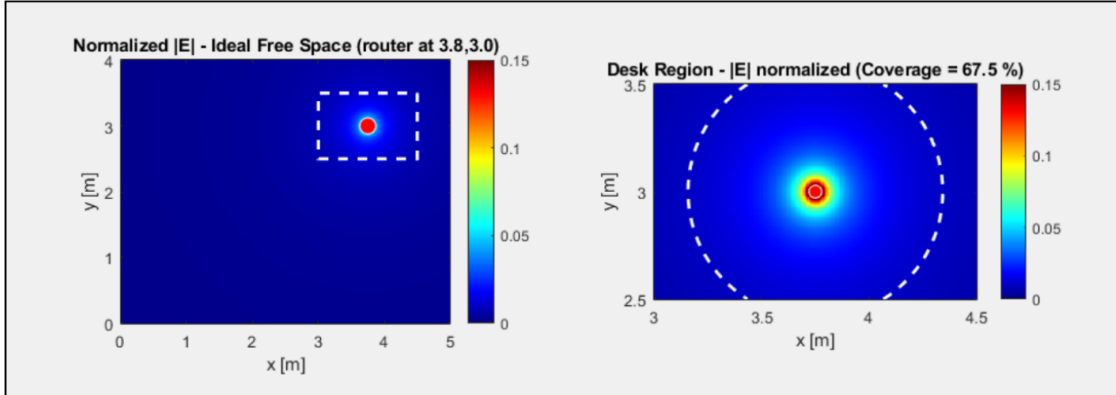


Figure 4: Ideal free-space reference: normalized field magnitude in the room and desk region with router at desk center and about 68% desk coverage.

The left plot shows the normalized free-space field in the full 5 by 4 m room, with the router at the desk center and colors smoothly fading from red to blue in an almost circular pattern, without reflections or interference. The right plot zooms into the desk region, where the router appears as a red spot and the field weakens gradually towards the edges. The dashed white circle marks the good-signal threshold, and about 67.5% of the desk area lies inside it, meaning roughly two-thirds of the desk is well covered under ideal conditions.

7.3 Reflective Walls

To model fully reflective walls, we change the system's boundary conditions to treat them like perfect conductors: the field is forced to vanish at the boundary, so every wave that reaches a wall is reflected back into the room. The cabinet is also modeled as a perfect conductor, so it behaves like a hard metallic obstacle that produces strong local reflections. The interior of the room is still free space where the time-harmonic Helmholtz problem is solved using the finite element method on a triangular mesh, with a small Gaussian source placed at each of the four candidate transmitter locations in turn. As a result, the room behaves almost like a cavity: waves bounce back and forth many times, and the steady-state field is the sum of a very large number of multipath contributions. This behavior is clearly visible in the four normalized field maps for the reflective case, where the color patterns are extremely fine grained and almost speckled across the whole room. Each transmitter position produces a dense standing wave pattern of bright and dark stripes that cover both the main propagation direction and the side walls. The dashed white rectangle still marks the desk region, but it is much harder to visually identify a single dominant beam because energy has been redistributed throughout the room.

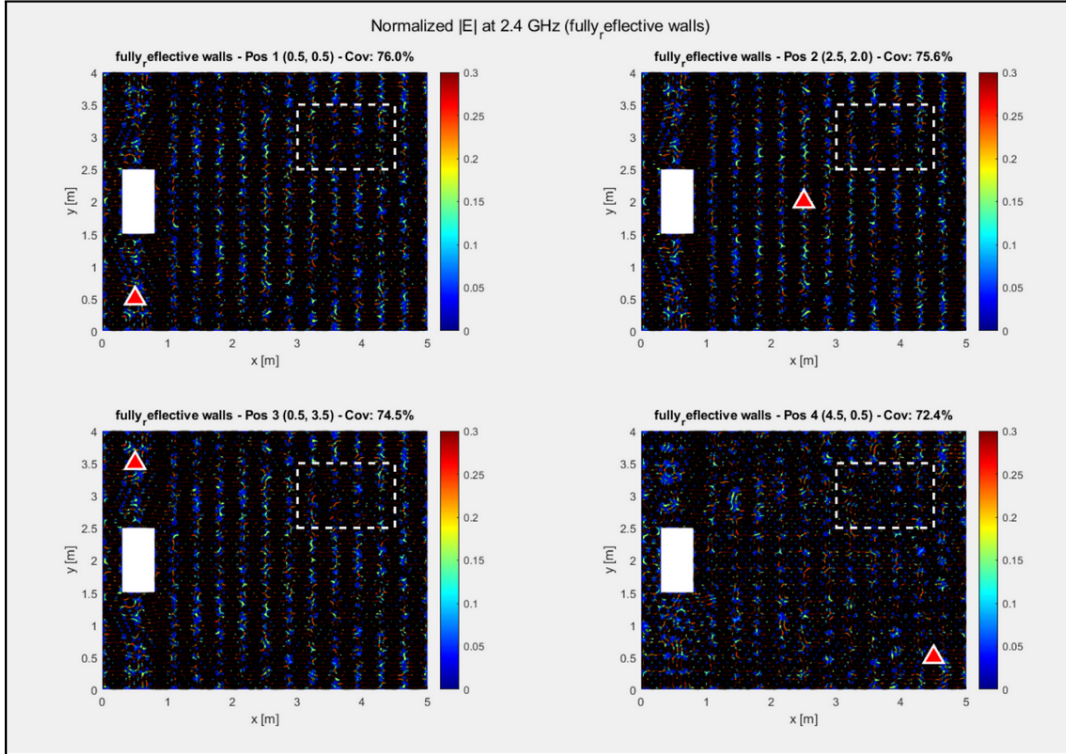


Figure 5: Field coverage for reflective walls.

The coverage bar chart for reflective walls shows that, despite the visually noisy pattern, the desk actually receives fairly robust signals in all four configurations. The coverage percentages cluster around seventy to seventy six percent for the chosen ten percent threshold. This means that most of the desk area sits above the chosen normalized field level, almost regardless of where the router is placed. The reason is that the reflective boundaries recycle power that would otherwise leave the room, so destructive interference in one part of the desk is usually compensated by constructive interference somewhere else nearby. When this behavior is compared to the ideal analytical free-

space solution, where the router sits at the center of the desk and the field decays monotonically like one over distance with no reflections, the fully reflective case looks slightly worse visually but comparable or even better in terms of average coverage.

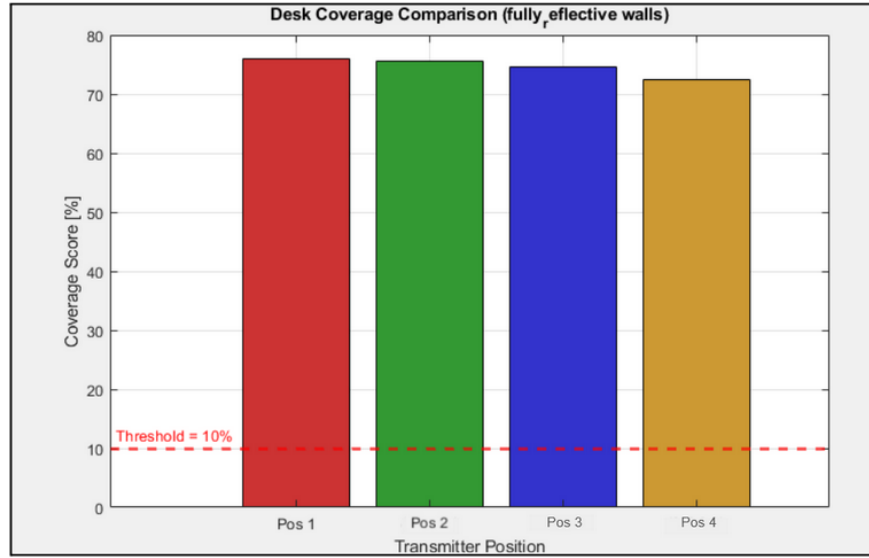


Figure 6: Field coverage for reflective walls. Threshold value $T_v = 10\%$.

7.4 Partially Absorbing Walls

To model the partially absorbing walls, we solve the Helmholtz equation on the same mesh with the router modeled as a Gaussian source at the same four positions, and an impedance-type condition is used that approximates a radiation condition. Physically, this represents a wall that lets part of the energy leak out of the room instead of reflecting it perfectly. As a result, the room is no longer a strict cavity. Waves still reflect from the walls because the impedance does not match free space exactly, but on each bounce some power is lost, so after a few trips the energy essentially leaves the system. The absorbing wall field maps show much clearer beam-like structures and fewer fine oscillations than the reflective ones. For each transmitter position one can visually trace a main lobe that travels away from the router and then gradually fades as it hits the walls or is shadowed by the cabinet. The standing wave pattern is still present, particularly close to the source and near the cabinet, but the fringes are more spread out and the field eventually decays toward the corners.

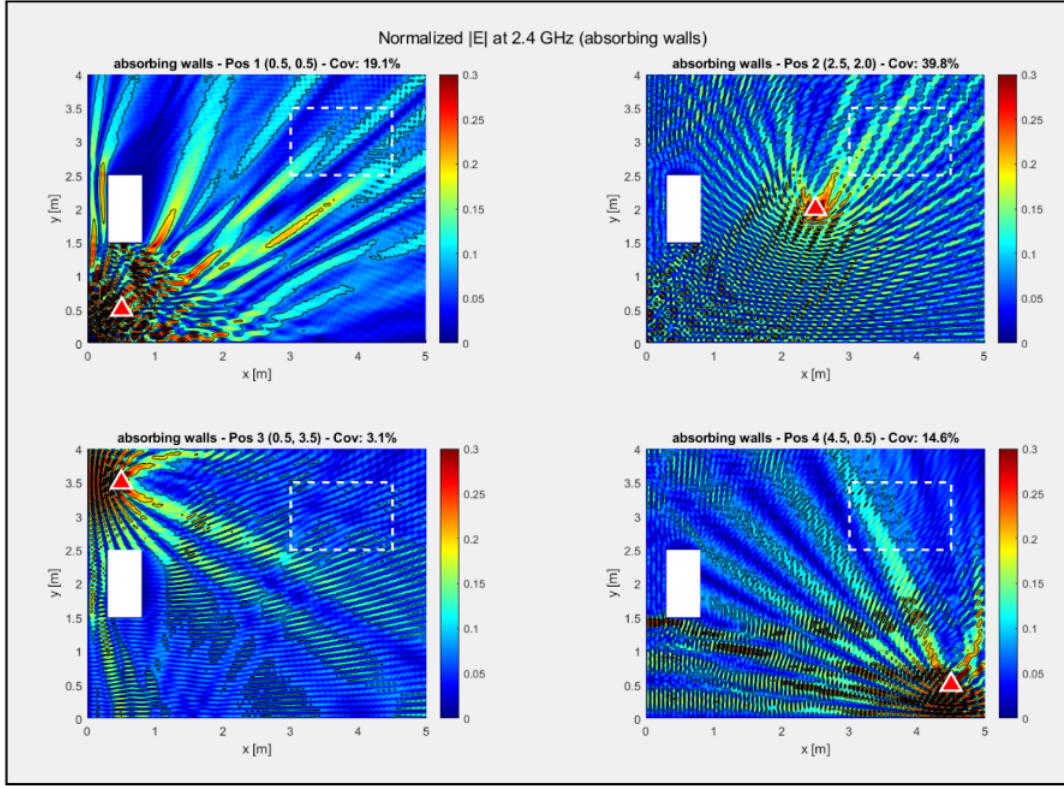


Figure 7: Field coverage for partially absorbing walls.

The absorbing coverage bar chart immediately shows the downside of letting energy leave the room. For the same ten percent threshold the coverage percentages drop dramatically and become very sensitive to transmitter placement: position two near the center of the room reaches around forty percent coverage, whereas position three near the upper left wall barely clears three percent. Positions one and four sit in between but still below twenty percent. This difference lines up with the field maps, where only the centrally placed router sends a strong beam into the desk region while the others either point away or are heavily distorted by interactions with the cabinet and the partially absorbing walls. The detailed best-position figure for absorbing walls makes the mechanism very clear.

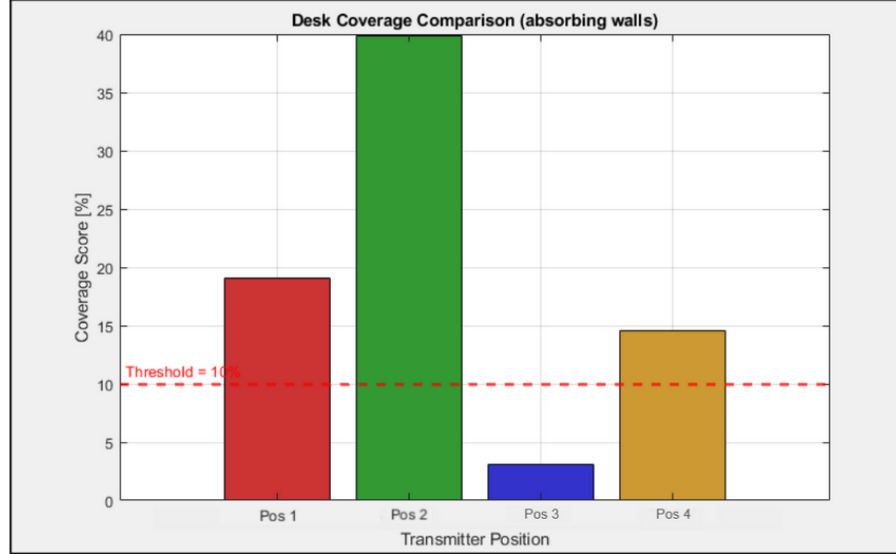


Figure 8: Field coverage for partially absorbing walls.

7.5 Threshold Value

Lowering the coverage threshold from 10 percent to 5 percent effectively relaxes the definition of what counts as good signal, and we compared the coverage of the partially reflective walls with the fully reflective case. In both models the field on the desk has a wide spread of values: a small region near the router sits close to the maximum, many points sit in the mid-range, and only a fraction fall into very deep nulls. With the higher 10 percent threshold, coverage is decided by how much of the desk lies above a demanding cut, so the fully reflective room looks much better because multiple reflections create strong constructive peaks that push large areas well above that level, while the absorbing room loses energy at the walls and has fewer strong peaks, so many mid-level points fall just below the threshold and are counted as uncovered. Once the threshold is lowered to 5 percent, many of those mid-level points in the absorbing case suddenly qualify as covered, because they were already between 5 and 10 percent of the local maximum field. The reflective case also gains a little coverage when the threshold moves down, but it was already covering most of the desk at 10 percent, so there are fewer new points to add. The net effect is that the absorbing bars move up more than the reflective bars, which pulls the two sets of bars much closer together.

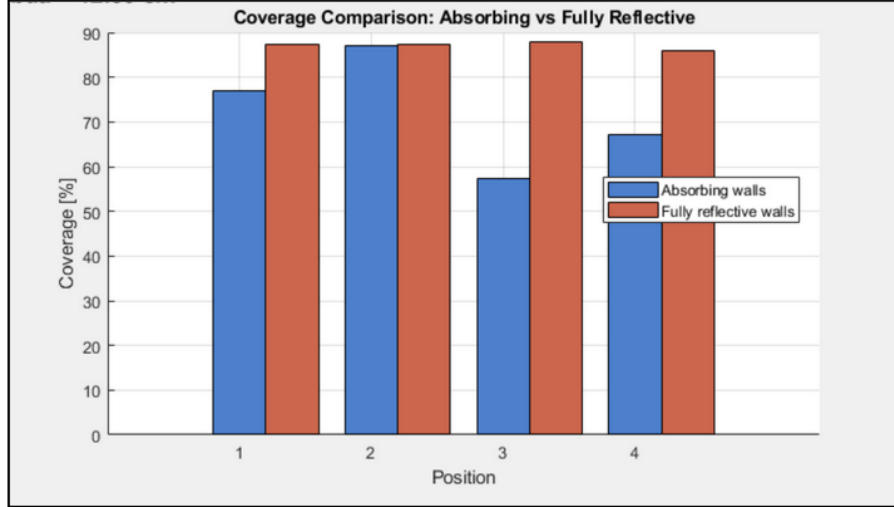


Figure 9: At value of 5%.

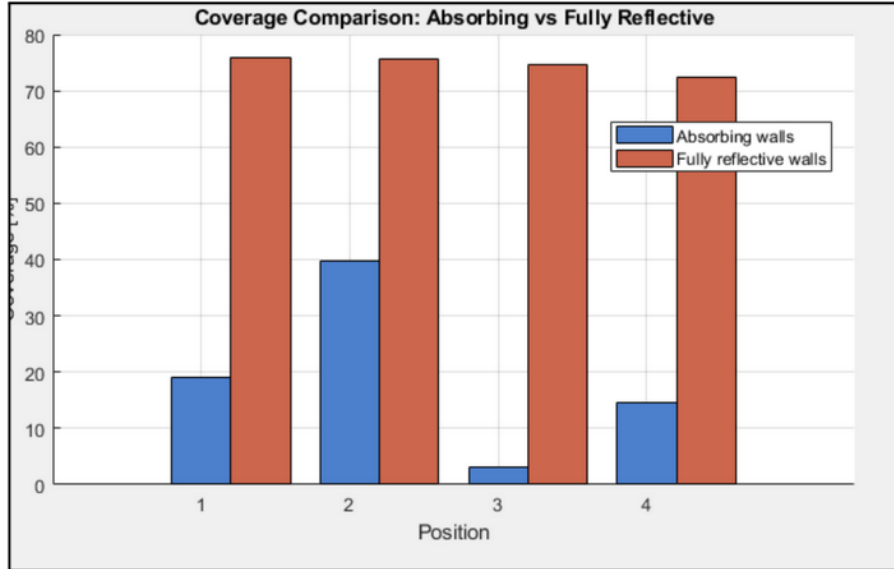


Figure 10: At threshold value of 10%.

7.6 Optimal Location, Limitations, and Future Improvements

In the end, the most realistic picture of an indoor WiFi link comes from the partially absorbing wall model with the router placed near the center of the room, close to position 2 in the figures. Real walls are neither perfect mirrors nor completely transparent: drywall, concrete, and glass all reflect part of the field and absorb the rest, while energy also leaks through doors and windows. The absorbing boundary condition used in the code mimics exactly that situation by letting waves leave the computational domain instead of trapping them forever as in the fully reflective case. When the coverage maps for absorbing walls are read together with the bar chart, it becomes clear that the central transmitter gives the best compromise between direct line of sight to the desk and a reasonably uniform standing wave pattern across the room. Positions 1, 3, and 4 either sit in

corners or near obstacles, so their beams skim the desk or are chopped up by strong interactions with the cabinet. Position 2 pushes energy into the upper right quadrant where the desk sits, and the standing-wave plots for this case show that, while the field still oscillates around the coverage threshold, the average level at desk height is higher than for the other locations. Compared to the ideal analytical free-space solution and the extreme cavity-like reflective case, the absorbing central-router scenario sits in the middle: it does not achieve the almost cavity-boosted coverage of fully reflective walls, but it avoids the unrealistic assumption that nothing ever leaves the room, and it shows how much coverage drops once realistic losses and leakage are taken seriously.

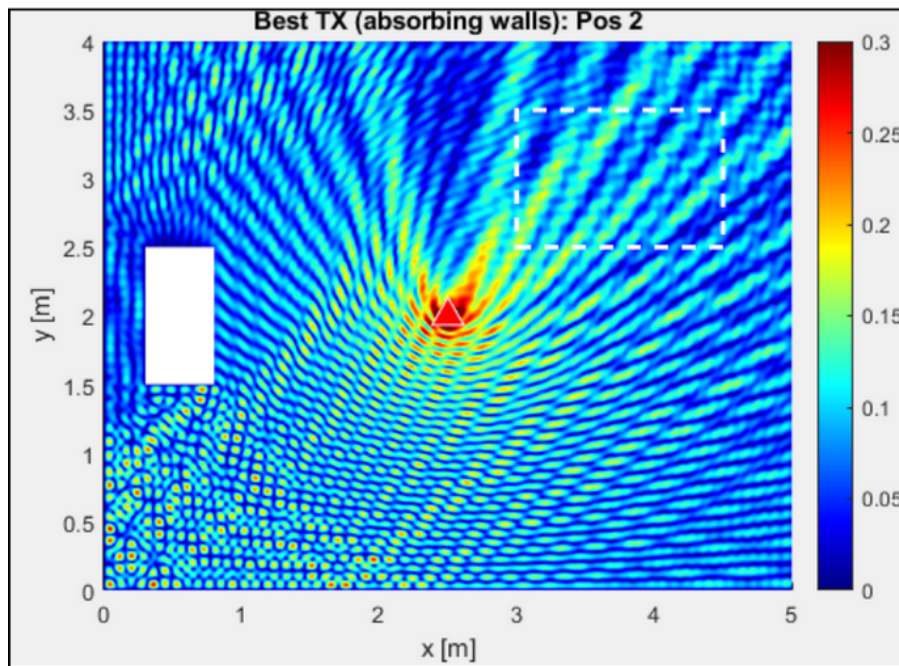


Figure 11: Ideal position on given parameters, and realistic boundary and initial conditions chosen.

A key design parameter in this whole study is the coverage threshold, which is set as a fixed fraction of the maximum normalized field in the room and then drawn as a horizontal red line in the bar plots and line profiles. That threshold implicitly stands for the minimum field level that would still support the target data rate, something like the familiar two-bars of WiFi point on a phone. If the threshold is raised, for example to demand only very strong fields, the reported coverage percentages would fall sharply and the desk would appear to have many more dead zones, even in the reflective case. If the threshold is lowered, on the other hand, the reported coverage improves on paper because more desk points are counted as good signals, but those extra areas would correspond to weaker links with lower throughput and higher sensitivity to noise and interference. The figures already show this behavior: the field profiles through the desk region for absorbing walls sit just above and below the current threshold, so nudging that level up or down would move a big chunk of the desk from red to green or vice versa. In a more refined design study the threshold would not be an arbitrary percentage, but would be tied to a received power level or signal-to-noise ratio derived from link budget calculations, so adjusting it would correspond directly to choosing between strong but smaller coverage and weaker but larger coverage.

Taken as a whole, the project gives a very intuitive and visually rich story of how a single WiFi router illuminates a real room, but it should not be mistaken for a perfect prediction engine. Several

simplifications limit quantitative accuracy. The room is modeled as a flat 2D slice at desk height, so the vertical variation of the field between floor and ceiling is ignored and the ceiling is only felt indirectly through the boundary condition. The electromagnetic problem is reduced to a scalar Helmholtz equation in a single polarization, whereas a real access point uses vector fields, dual polarization, and often multiple antennas. The outer walls are represented only through effective boundary conditions, not as finite thickness layers with frequency dependent permittivity and loss, and the rest of the furniture and human bodies are completely absent from the geometry. The router itself is idealized as an isotropic Gaussian source with fixed amplitude, so real antenna patterns, transmit power control, and hardware details are not captured. Finally, the model works at a single frequency and in a steady state regime, while real WiFi uses wideband signals, time varying channels, and interference from other networks. All of these choices make the simulations fast and interpretable, but they also mean that the numbers on the plots should be read as trends and relative comparisons rather than as exact predictions in dBm.

These limitations suggest natural directions for improvement if the method were to be pushed toward engineering design. A next step would be to move from a 2D slice to a full 3D model of the room, including the floor, ceiling, and vertical placement of the router and receivers, and to assign more realistic material properties to walls, doors, windows, and the metal cabinet using measured permittivity and loss tangent values. Additional geometry such as chairs, tables, and even simplified human phantoms could be added to see how occupancy changes the standing wave patterns. On the numerical side, adaptive mesh refinement could concentrate elements where the field gradients are strongest, and the solver could be run at several nearby frequencies to mimic a small portion of the WiFi band instead of a single tone. Most importantly, the model could be calibrated against real measurements in the same room, so that the boundary conditions and source representation are tuned until the simulated coverage fields line up with what a laptop or phone actually sees. With those improvements, the same finite element framework used here would move from being a very informative teaching tool to a more robust design aid, while still keeping the core idea alive: start from the underlying Helmholtz physics, build a realistic domain, and then read router placement and coverage decisions directly from the resulting field maps and standing wave patterns.

7.7 Innovation Opportunities

Now that we have used PDEs to determine the most optimal transmitter location in a room, we can discuss the subsequent innovation opportunities behind this model. Since the Helmholtz equation is frequency-dependent, by sweeping frequencies and polarizations within the PDE model, we can determine which frequency bands offer the highest overall coverage across multiple transmitter locations. We can also develop AI-driven solution approaches that simplify the model and the simulation process. We can perform PDE-constrained optimization because the Helmholtz problem is well-posed and differentiable, and use gradient-based methods such as adjoint PDEs and automatic differentiation to optimize AP position, orientation, or even wall impedance subject to coverage constraints. Moreover, we can use the current PDE model to develop Physics-Informed Neural Networks, or hybrid models, by embedding the Helmholtz equation as a soft constraint in a neural network loss. A PINN is a neural network that learns the field pattern by taking the spatial coordinates as input and being trained on data while satisfying the Helmholtz PDE and its boundary conditions through its loss function. On the boundaries, the network penalizes any deviation from the boundary conditions, and it optionally matches measured data points. The

result is a mesh-free, continuous surrogate that approximates the PDE solution everywhere, can incorporate sparse measurements, and is fast to evaluate for many design configurations.

Our PDE model could also be extended to a larger scale, focusing on Free-Space Optical systems and rooftop-to-rooftop RF, mmWave, or THz links. Since light is an electromagnetic wave, it satisfies the Helmholtz equation with a wavenumber for its high frequency, meaning that we can model optical communication systems with the same PDE framework. To effectively model the FSO wireless link, which is a rooftop-to-rooftop transmission of a laser, we rely on the Helmholtz equation along with the Gaussian source model of the transmitter to determine the optical spatial pattern of the light being transmitted. By studying the different optical field patterns, we can determine the optimal location of the optical aperture to allow for the strongest coverage at the receiver. To model FSO links and large-scale rooftop-to-rooftop RF, mmWave, or THz links, we extend the PDE domain to 3D to accurately model houses and buildings, while incorporating boundary conditions to represent reflection, absorption, or scattering from different materials.

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